

Plane Geometry

ANSWER KEY



Angle relationships

- 1 Two angles are supplementary, and one is 3 times the other. Find both angles.

Let the angles be x and $3x$: $x + 3x = 180$, so $4x = 180$, giving $x = 45$. The angles are 45° and 135° .

- 2 Two parallel lines are cut by a transversal, creating an angle of 70° . Find the measure of its alternate interior angle and its corresponding angle.

Alternate interior angles are equal: 70° . Corresponding angles are also equal: 70° .

Triangle angle sum and exterior angles

- 3 A triangle has angles measuring x , $2x$, and $3x$. Find each angle.

$x + 2x + 3x = 180$, so $6x = 180$, giving $x = 30$. Angles: 30° , 60° , 90° .

- 4 A triangle's exterior angle at one vertex measures 115° . If one of the remote interior angles is 50° , find the other remote interior angle.

By the exterior angle theorem, the exterior angle equals the sum of the two remote interior angles:
 $115 = 50 + x$, so $x = 65^\circ$.

Area and perimeter of polygons

- 5 Find the area of a triangle with base 14 and height 9.

$$\text{Area} = \frac{1}{2}(14)(9) = 63.$$

- 6 Find the area of a parallelogram with base 12 and height 5.

$$\text{Area} = \text{base} \times \text{height} = 12 \times 5 = 60.$$

- 7 Find the area of a regular hexagon with side length 6, using the formula $A = \frac{3\sqrt{3}}{2}s^2$.

$$A = \frac{3\sqrt{3}}{2}(6)^2 = \frac{3\sqrt{3}}{2}(36) = 54\sqrt{3}.$$

The Pythagorean theorem and special right triangles

- 8 A right triangle has legs 9 and 12. Find the hypotenuse.

$$c = \sqrt{9^2 + 12^2} = \sqrt{81 + 144} = \sqrt{225} = 15.$$

- 9 A 45-45-90 triangle has a leg of length 8. Find the length of the hypotenuse.

In a 45-45-90 triangle, the hypotenuse equals leg $\times \sqrt{2}$: $8\sqrt{2}$.

- 10 A 30-60-90 triangle has a hypotenuse of 20. Find the lengths of both legs.

In a 30-60-90 triangle, the shorter leg is $\frac{\text{hypotenuse}}{2}$ and the longer leg is $\frac{\text{hypotenuse}}{2} \times \sqrt{3}$: shorter leg = 10, longer leg = $10\sqrt{3}$.

Circles: area, circumference, arcs, and sectors

- 11 Find the area and circumference of a circle with diameter 14, in terms of π .

Radius = 7. Area = $\pi r^2 = 49\pi$. Circumference = $\pi d = 14\pi$.

- 12 Find the arc length of a 120° sector in a circle with radius 9, in terms of π .

Arc length = $\frac{120}{360} \times 2\pi(9) = \frac{1}{3} \times 18\pi = 6\pi$.

Similar and congruent figures

- 13 Two similar triangles have a scale factor of $\frac{3}{5}$. If the smaller triangle has area 18, find the area of the larger triangle.

Areas of similar figures scale by the SQUARE of the linear scale factor: $\left(\frac{5}{3}\right)^2 = \frac{25}{9}$. Larger triangle's area: $18 \times \frac{25}{9} = 50$.

Volume and surface area of 3D solids

- 14 Find the volume of a rectangular prism with dimensions 5, 8, and 10.

Volume = $5 \times 8 \times 10 = 400$.

- 15 Find the volume of a cone with radius 6 and height 10, in terms of π .

Volume = $\frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(36)(10) = 120\pi$.

Multi-step geometry problems

- 16** This question has two parts. A square garden has area 144 square feet. (a) Find the side length and perimeter of the garden. (b) A circular fountain with radius 3 feet is placed inside the garden. Find the remaining garden area (not covered by the fountain), in terms of π .

(a) Side length: $\sqrt{144} = 12$ feet. Perimeter: $4(12) = 48$ feet. (b) Fountain area: $\pi(3)^2 = 9\pi$. Remaining area: $144 - 9\pi$ square feet.

Trapezoids and composite polygon areas

- 17** Find the area of a trapezoid with parallel sides 8 and 14, and height 6.

$$\text{Area} = \frac{1}{2}(8 + 14)(6) = \frac{1}{2}(22)(6) = 66.$$

- 18** An L-shaped figure is formed by a 10-by-8 rectangle with a 4-by-3 rectangle removed from one corner. Find the remaining area.

Large rectangle area: $10 \times 8 = 80$. Removed rectangle area: $4 \times 3 = 12$. Remaining area: $80 - 12 = 68$.

Chords, tangents, and inscribed angles in circles

- 19** An inscribed angle subtends an arc of 100° . Find the measure of the inscribed angle.

An inscribed angle is always half the arc it subtends: $\frac{100^\circ}{2} = 50^\circ$.

- 20** A tangent line touches a circle at point P . Explain the relationship between the tangent line and the radius drawn to point P .

A tangent line is always perpendicular to the radius drawn to the point of tangency — this is a fundamental property relating tangent lines and circles.

Congruence and triangle proofs

- 21** State the triangle congruence criterion (SSS, SAS, ASA, AAS) that applies when two triangles share two equal angles and the included side.

ASA (Angle-Side-Angle) applies, since two angles and the side directly between them are all given as congruent.

A multi-part composite geometry problem

- 22 This question has two parts. A semicircular window has a diameter of 12 feet and sits atop a rectangular window of width 12 feet and height 5 feet. (a) Find the area of the semicircular portion, in terms of π . (b) Find the total area of the combined window shape.
- (a) Radius = 6. Semicircle area = $\frac{1}{2}\pi r^2 = \frac{1}{2}\pi(36) = 18\pi$. (b) Rectangle area: $12 \times 5 = 60$. Total area: $60 + 18\pi$ square feet.
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Multiple choice — easy

- 23 Two angles are complementary, and they differ by 52° . Find the larger angle.

- a. 19°
- b. 66°
- c. 76°
- d. 71° ✓

Let the angles be x and $x + 52$: $x + (x + 52) = 90$, so $2x = 38$, giving $x = 19$. The larger angle is $19 + 52 = 71^\circ$.

- 24 Two angles are complementary, and they differ by 14° . Find the larger angle.

- a. 52° ✓
- b. 38°
- c. 47°
- d. 57°

Let the angles be x and $x + 14$: $x + (x + 14) = 90$, so $2x = 76$, giving $x = 38$. The larger angle is $38 + 14 = 52^\circ$.

25 Two angles are complementary, and they differ by 8° . Find the larger angle.

a. 49° ✓

b. 44°

c. 41°

d. 54°

Let the angles be x and $x + 8$: $x + (x + 8) = 90$, so $2x = 82$, giving $x = 41$. The larger angle is $41 + 8 = 49^\circ$.

26 Two angles are supplementary, and they differ by 4° . Find the larger angle.

a. 87°

b. 88°

c. 97°

d. 92° ✓

Let the angles be x and $x + 4$: $x + (x + 4) = 180$, so $2x = 176$, giving $x = 88$. The larger angle is $88 + 4 = 92^\circ$.

27 Two angles are complementary, and they differ by 48° . Find the larger angle.

a. 74°

b. 64°

c. 21°

d. 69° ✓

Let the angles be x and $x + 48$: $x + (x + 48) = 90$, so $2x = 42$, giving $x = 21$. The larger angle is $21 + 48 = 69^\circ$.

28 Two angles are supplementary, and they differ by 14° . Find the larger angle.

- a. 92°
- b. 102°
- c. 83°
- d. 97° ✓

Let the angles be x and $x + 14$: $x + (x + 14) = 180$, so $2x = 166$, giving $x = 83$. The larger angle is $83 + 14 = 97^\circ$.

29 Two angles are supplementary, and they differ by 46° . Find the larger angle.

- a. 67°
- b. 108°
- c. 113° ✓
- d. 118°

Let the angles be x and $x + 46$: $x + (x + 46) = 180$, so $2x = 134$, giving $x = 67$. The larger angle is $67 + 46 = 113^\circ$.

30 Two angles are complementary, and they differ by 32° . Find the larger angle.

- a. 61° ✓
- b. 56°
- c. 29°
- d. 66°

Let the angles be x and $x + 32$: $x + (x + 32) = 90$, so $2x = 58$, giving $x = 29$. The larger angle is $29 + 32 = 61^\circ$.

31 Two angles are supplementary, and they differ by 30° . Find the larger angle.

a. 105° ✓

b. 100°

c. 75°

d. 110°

Let the angles be x and $x + 30$: $x + (x + 30) = 180$, so $2x = 150$, giving $x = 75$. The larger angle is $75 + 30 = 105^\circ$.

32 Two angles are complementary, and they differ by 34° . Find the larger angle.

a. 62° ✓

b. 67°

c. 57°

d. 28°

Let the angles be x and $x + 34$: $x + (x + 34) = 90$, so $2x = 56$, giving $x = 28$. The larger angle is $28 + 34 = 62^\circ$.

33 Two angles are complementary, and they differ by 22° . Find the larger angle.

a. 34°

b. 61°

c. 51°

d. 56° ✓

Let the angles be x and $x + 22$: $x + (x + 22) = 90$, so $2x = 68$, giving $x = 34$. The larger angle is $34 + 22 = 56^\circ$.

34 Two angles are supplementary, and they differ by 130° . Find the larger angle.

a. 155° ✓

b. 25°

c. 160°

d. 150°

Let the angles be x and $x + 130$: $x + (x + 130) = 180$, so $2x = 50$, giving $x = 25$. The larger angle is $25 + 130 = 155^\circ$.

35 Two angles are complementary, and they differ by 76° . Find the larger angle.

a. 88°

b. 7°

c. 78°

d. 83° ✓

Let the angles be x and $x + 76$: $x + (x + 76) = 90$, so $2x = 14$, giving $x = 7$. The larger angle is $7 + 76 = 83^\circ$.

36 Two angles are supplementary, and they differ by 170° . Find the larger angle.

a. 175° ✓

b. 170°

c. 5°

d. 180°

Let the angles be x and $x + 170$: $x + (x + 170) = 180$, so $2x = 10$, giving $x = 5$. The larger angle is $5 + 170 = 175^\circ$.

37 Two angles are complementary, and they differ by 24° . Find the larger angle.

a. 57° ✓

b. 52°

c. 33°

d. 62°

Let the angles be x and $x + 24$: $x + (x + 24) = 90$, so $2x = 66$, giving $x = 33$. The larger angle is $33 + 24 = 57^\circ$.

Multiple choice — medium

38 A right triangle has legs 12 and 16. Find the length of the hypotenuse.

a. 28

b. 21

c. 19

d. 20 ✓

$$c = \sqrt{12^2 + 16^2} = \sqrt{144 + 256} = \sqrt{400} = 20.$$

39 A right triangle has legs 16 and 30. Find the length of the hypotenuse.

a. 35

b. 34 ✓

c. 33

d. 46

$$c = \sqrt{16^2 + 30^2} = \sqrt{256 + 900} = \sqrt{1156} = 34.$$

40 A right triangle has legs 27 and 36. Find the length of the hypotenuse.

- a. 63
- b. 44
- c. 46
- d. 45 ✓

$$c = \sqrt{27^2 + 36^2} = \sqrt{729 + 1296} = \sqrt{2025} = 45.$$

41 A right triangle has legs 6 and 8. Find the length of the hypotenuse.

- a. 9
- b. 11
- c. 14
- d. 10 ✓

$$c = \sqrt{6^2 + 8^2} = \sqrt{36 + 64} = \sqrt{100} = 10.$$

42 A right triangle has legs 40 and 96. Find the length of the hypotenuse.

- a. 105
- b. 136
- c. 104 ✓
- d. 103

$$c = \sqrt{40^2 + 96^2} = \sqrt{1600 + 9216} = \sqrt{10816} = 104.$$

43 A right triangle has legs 9 and 40. Find the length of the hypotenuse.

- a. 41 ✓
- b. 42
- c. 40
- d. 49

$$c = \sqrt{9^2 + 40^2} = \sqrt{81 + 1600} = \sqrt{1681} = 41.$$

44 A right triangle has legs 24 and 32. Find the length of the hypotenuse.

a. 40 ✓

b. 39

c. 41

d. 56

$$c = \sqrt{24^2 + 32^2} = \sqrt{576 + 1024} = \sqrt{1600} = 40.$$

45 A right triangle has legs 14 and 48. Find the length of the hypotenuse.

a. 50 ✓

b. 62

c. 49

d. 51

$$c = \sqrt{14^2 + 48^2} = \sqrt{196 + 2304} = \sqrt{2500} = 50.$$

46 A right triangle has legs 15 and 36. Find the length of the hypotenuse.

a. 38

b. 40

c. 51

d. 39 ✓

$$c = \sqrt{15^2 + 36^2} = \sqrt{225 + 1296} = \sqrt{1521} = 39.$$

47 A right triangle has legs 10 and 24. Find the length of the hypotenuse.

a. 27

b. 26 ✓

c. 25

d. 34

$$c = \sqrt{10^2 + 24^2} = \sqrt{100 + 576} = \sqrt{676} = 26.$$

48 A right triangle has legs 7 and 24. Find the length of the hypotenuse.

a. 31

b. 24

c. 25 ✓

d. 26

$$c = \sqrt{7^2 + 24^2} = \sqrt{49 + 576} = \sqrt{625} = 25.$$

49 A right triangle has legs 18 and 24. Find the length of the hypotenuse.

a. 42

b. 31

c. 30 ✓

d. 29

$$c = \sqrt{18^2 + 24^2} = \sqrt{324 + 576} = \sqrt{900} = 30.$$

50 A right triangle has legs 36 and 48. Find the length of the hypotenuse.

a. 84

b. 60 ✓

c. 59

d. 61

$$c = \sqrt{36^2 + 48^2} = \sqrt{1296 + 2304} = \sqrt{3600} = 60.$$

51 A right triangle has legs 20 and 48. Find the length of the hypotenuse.

a. 52 ✓

b. 68

c. 53

d. 51

$$c = \sqrt{20^2 + 48^2} = \sqrt{400 + 2304} = \sqrt{2704} = 52.$$

52 A right triangle has legs 80 and 84. Find the length of the hypotenuse.

a. 115

b. 117

c. 116 ✓

d. 164

$$c = \sqrt{80^2 + 84^2} = \sqrt{6400 + 7056} = \sqrt{13456} = 116.$$

Multiple choice — hard

53 A circle has radius 11. Find the arc length of a 270° sector, in terms of π .

a. $\frac{33}{2}\pi$ ✓

b. $\frac{35}{2}\pi$

c. $\frac{31}{2}\pi$

d. 33π

$$\text{Arc length} = \frac{\theta}{360^\circ} \times 2\pi r = \frac{270}{360} \times 2\pi(11) = \frac{33}{2}\pi.$$

54 A circle has radius 9. Find the arc length of a 120° sector, in terms of π .

- a. 5π
- b. 6π ✓
- c. 12π
- d. 7π

$$\text{Arc length} = \frac{\theta}{360^\circ} \times 2\pi r = \frac{120}{360} \times 2\pi(9) = 6\pi.$$

55 A circle has radius 2. Find the sector area of a 180° sector, in terms of π .

- a. 2π ✓
- b. 3π
- c. 4π
- d. 1π

$$\text{Sector area} = \frac{\theta}{360^\circ} \times \pi r^2 = \frac{180}{360} \times \pi(4) = 2\pi.$$

56 A circle has radius 12. Find the sector area of a 180° sector, in terms of π .

- a. 144π
- b. 71π
- c. 73π
- d. 72π ✓

$$\text{Sector area} = \frac{\theta}{360^\circ} \times \pi r^2 = \frac{180}{360} \times \pi(144) = 72\pi.$$

57 A circle has radius 12. Find the arc length of a 30° sector, in terms of π .

- a. 1π
- b. 3π
- c. 2π ✓
- d. 4π

$$\text{Arc length} = \frac{\theta}{360^\circ} \times 2\pi r = \frac{30}{360} \times 2\pi(12) = 2\pi.$$

58 Find the volume of a rectangular prism with dimensions 10, 9, and 7.

- a. 446
- b. 620
- c. 640
- d. 630 ✓

$$\text{Volume} = 10 \times 9 \times 7 = 630.$$

59 Find the volume of a cylinder with radius 5 and height 14, in terms of π .

- a. 350π ✓
- b. 360π
- c. 140π
- d. 340π

$$\text{Volume} = \pi r^2 h = \pi(25)(14) = 350\pi.$$

60 Find the volume of a cylinder with radius 2 and height 15, in terms of π .

- a. 50π
- b. 70π
- c. 60π ✓
- d. 56π

$$\text{Volume} = \pi r^2 h = \pi(4)(15) = 60\pi.$$

61 Find the volume of a cone with radius 5 and height 10, in terms of π .

a. $\frac{244}{3}\pi$

b. $\frac{250}{3}\pi$ ✓

c. $\frac{256}{3}\pi$

d. 250π

$$\text{Volume} = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(25)(10) = \frac{250}{3}\pi.$$

62 Find the volume of a cone with radius 8 and height 10, in terms of π .

a. $\frac{634}{3}\pi$

b. 640π

c. $\frac{646}{3}\pi$

d. $\frac{640}{3}\pi$ ✓

$$\text{Volume} = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(64)(10) = \frac{640}{3}\pi.$$

63 Two similar triangles have a scale factor of $\frac{2}{5}$. If the smaller triangle has area 16, find the area of the larger triangle.

a. 105

b. 40

c. 95

d. 100 ✓

Areas of similar figures scale by the square of the linear scale factor: $\left(\frac{5}{2}\right)^2 = \frac{25}{4}$. Larger triangle's area: $16 \times 25/4 = 100$.

- 64 Two similar triangles have a scale factor of $\frac{1}{3}$. If the smaller triangle has area 1, find the area of the larger triangle.

a. 9 ✓

b. 3

c. 4

d. 14

Areas of similar figures scale by the square of the linear scale factor: $\left(\frac{3}{1}\right)^2 = \frac{9}{1}$. Larger triangle's area: $1 \times 9/1 = 9$.

- 65 Two similar triangles have a scale factor of $\frac{1}{2}$. If the smaller triangle has area 1, find the area of the larger triangle.

a. 1

b. 9

c. 2

d. 4 ✓

Areas of similar figures scale by the square of the linear scale factor: $\left(\frac{2}{1}\right)^2 = \frac{4}{1}$. Larger triangle's area: $1 \times 4/1 = 4$.

- 66 Two similar triangles have a scale factor of $\frac{3}{5}$. If the smaller triangle has area 18, find the area of the larger triangle.

a. 50 ✓

b. 30

c. 45

d. 55

Areas of similar figures scale by the square of the linear scale factor: $\left(\frac{5}{3}\right)^2 = \frac{25}{9}$. Larger triangle's area: $18 \times 25/9 = 50$.

- 67 Two similar triangles have a scale factor of $\frac{1}{2}$. If the smaller triangle has area 2, find the area of the larger triangle.
- a. 3
 - b. 13
 - c. 4
 - d. 8 ✓

Areas of similar figures scale by the square of the linear scale factor: $\left(\frac{2}{1}\right)^2 = \frac{4}{1}$. Larger triangle's area: $2 \times 4/1 = 8$.